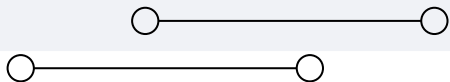


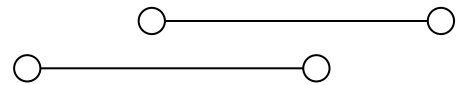


# WatMatch

**Creating Canada's Largest Capstone Ecosystem**

Siddharth, Samson, Ayman, Darsh, Sparsh

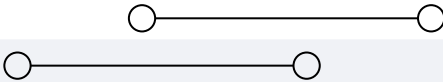




# Creating Canada's Largest Capstone Ecosystem

- Global Futures Initiative (more info [here](#))
- Our contacts:
  - Prof. Derek Rayside (Initiative Lead, Mentor)
  - Mr. Jamieson Cox (Centre for Work-Integrated Learning)
- We're first rolling out this initiative to UWaterloo students
- With iterative feedback, we plan to make this system accessible to all Canadian Universities





✦ 01 ✦

# Problem Recap



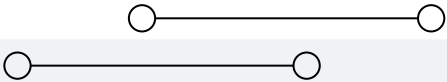
# Manual Matching Work

Current cross-faculty matching depends on large PDFs built from student survey responses

## 4 main problems

1. Students cannot easily search/compare opportunities
2. Most students form teams with people they already know
3. Faculty/staff coordinate matching by hand
4. No clear system of record for consent, approvals, or decisions for interdisciplinary matching





✦ 02 ✦

# Why this matters



# Who does this impact?

A platform for capstone matching helps...

## **Students**

Find teammates, projects, external opportunities beyond their own network!

## **Instructors**

Review proposals, give feedback, manage teams, track decisions in one centralized roster!

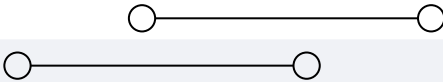
## **External Partners**

Share opportunities and ideas with students!

## **Universities**

To support interdisciplinary capstones at a large scale





03

# Current Status





# SE 390 Status



## Basic MVP Flow

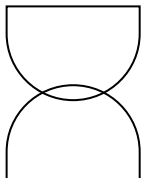
- Student logs in (+ updates basic profile)
- Student creates capstone proposal
  - Backend auto-creates team for student + links capstone to team
- Proposal appears in instructor's dashboard (can Accept/Reject)
- Only approved capstones appear in dashboard for other students to express interest in
- Role-aware login + dashboards for students & instructors

## Tech Stack

- Next.js frontend, FastAPI backend, Supabase for storage

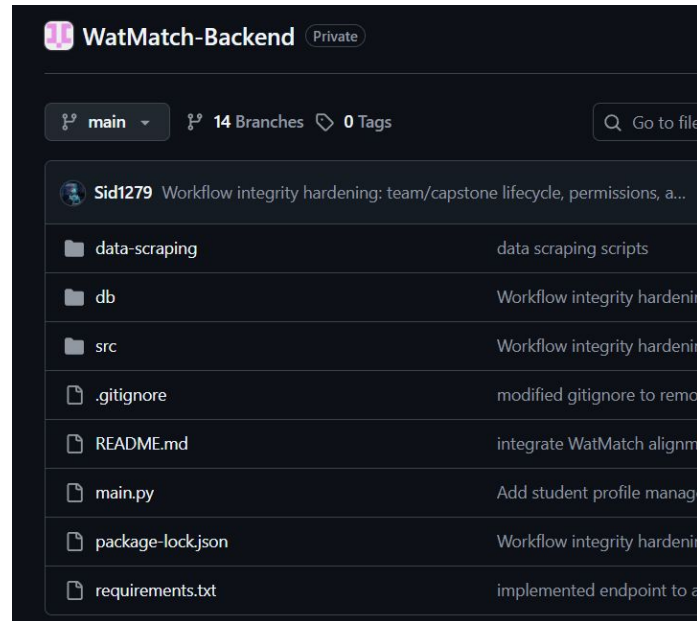
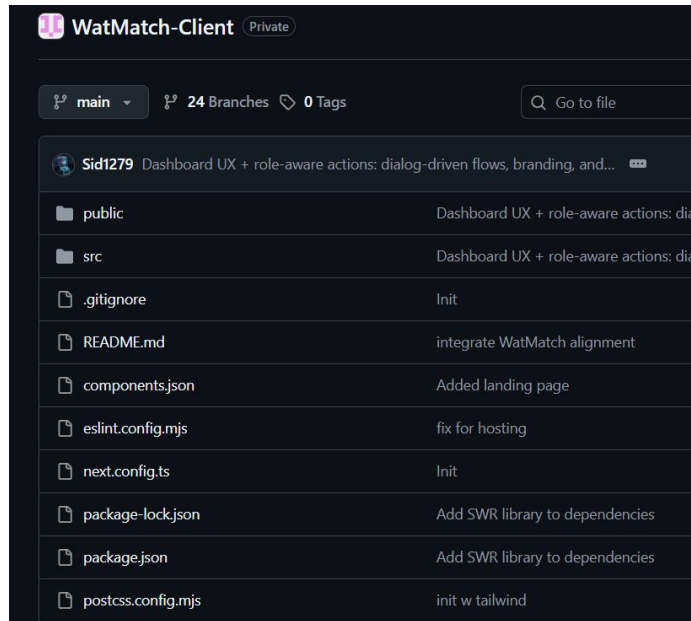
## Deployment Plan

- Put in a request for a VM from UW Engineering Computing
- Potential UW Portal Integration
- Prof. Rayside is in talks with IST & Engineering Computing to handle these requests

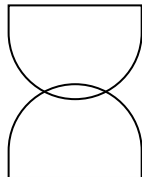




# SE 390 WatMatch GitHub Repos

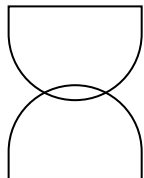
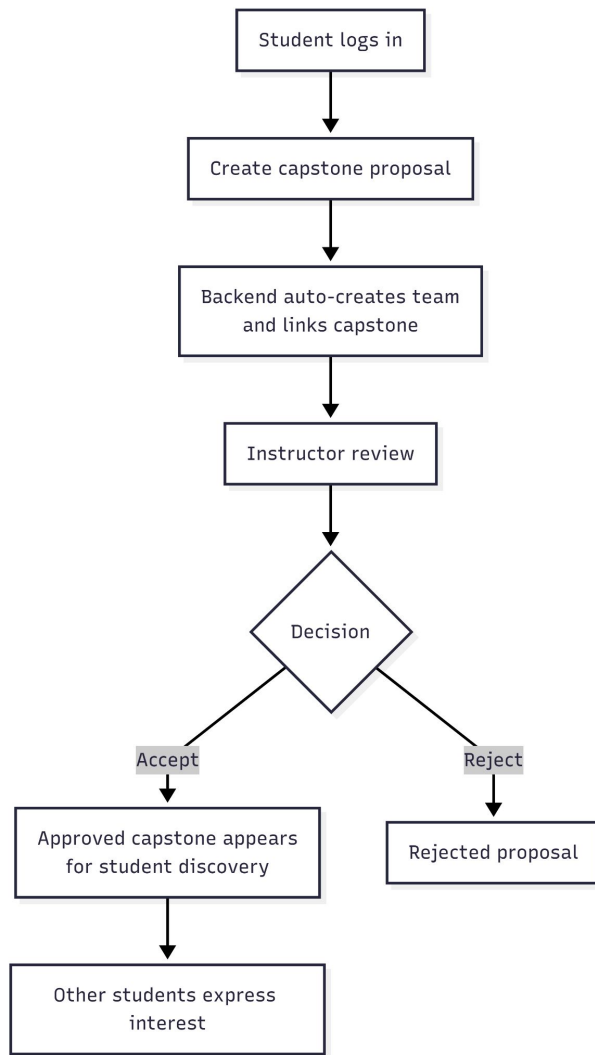


- Used GitHub Issues to implement features + resolve bugs
- Each PR needed 2+ reviews to be merged into main

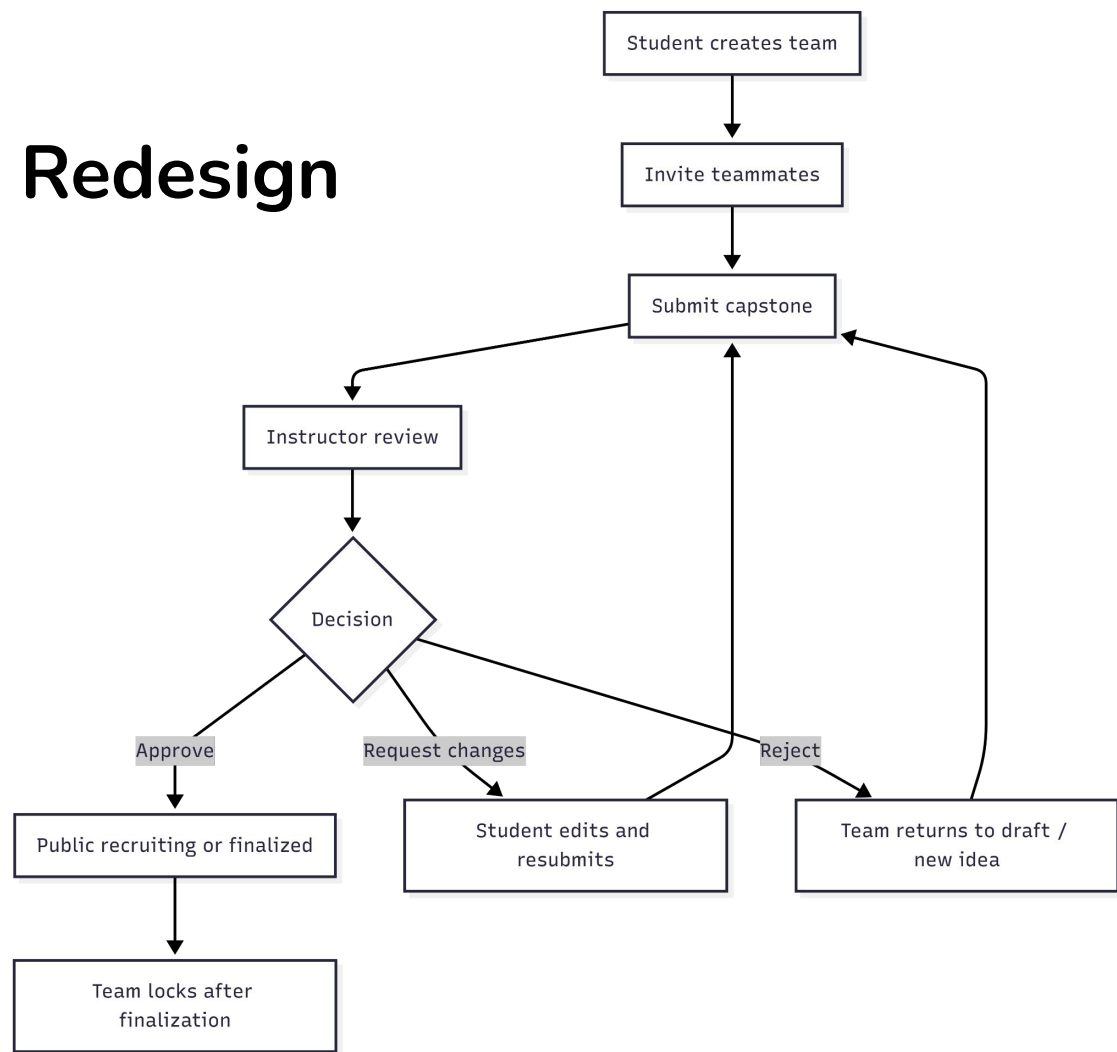


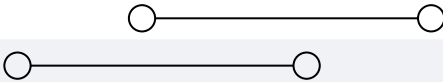


# SE 390 Design (Old)



# Post-SE 390 Redesign (New)





✦ 04 ✦

# Plan for Spring 2026





# SE 490 - Spring 2026

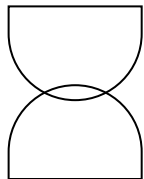


## Mid 1 - Foundation

- Schema redesign + migrate existing backend to Supabase RPCs for transactionality
- System Design diagrams (one per core flow)
- Feature #1: Students Browse Past Capstones (2016-2026)
- Feature #2: Admin Management (add/import courses, users to DB)

## Mid 2 - Single-Course WatMatch

- Feature #3: Let students create team without capstone
- Feature #4: Leaders can **invite** students to team; Students can express **interest** in capstones
- Feature #5: Instructor Accept/Reject/Request Changes
- Feature #6: Student's revision & resubmissions





# SE 490 - Spring 2026

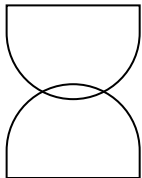


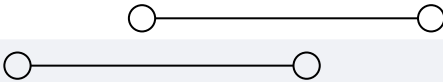
## **Final** - Multi-Course WatMatch with external partners

- Feature #7: Audit log for capstone lifecycle (instructor view only)
- Feature #8: Multi-course approval consensus
  - Suppose a student taking SE 490 wants to collaborate with students in ECE 498A
  - Instructor approval needed from BOTH courses
- Feature #9: External Partners post ideas; students can tag external partners in capstones
- Test for concurrent users, make all operations idempotent, no race conditions

## **Success Criteria** - Fully functional multi-course WatMatch (ready to deploy)

- Students complete team formation (invites & interests) + capstone submission cycles
- Instructors can review, accept, reject, or request changes to a capstone through the app
- Resolve any ambiguities for all proposed features/workflows by Mid 1





✦ 05 ✦

# Easy vs Risky Parts





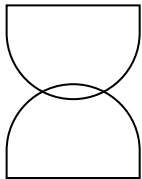
# Easy vs Risky Parts

## Easy Parts

- Past capstone data scraping + imports + pagination
- Admin course + user management
- UI cleanups + creating new dashboard & form templates
- Writing good documentation (for team in Fall to carry project forward)

## Risky Parts

- Multi-course approval rules
- Race conditions when two users act at the same time
- Invite and interest conflicts
- Team locking after instructor approval
- UW ADFS auth (assuming our mentor is able to get this sorted)





# Thank You

We'd love to hear your thoughts!

